Clinton Enwerem

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Summary

Robotics Research Engineer with 5+ years of research experience and Electrical and Computer Engineering Ph.D. candidate at the University of Maryland, graduating in Fall 2026. Develop demonstration data pipelines, evaluate imitation and reinforcement learning policies, and implement Python/C++ sensing and control software for dexterous manipulation in simulation and on physical, poly-articulated robots.

Skills

Robotics: ROS/ROS 2, ros2_control, MoveIt 2, MoveIt Servo, SocketCAN, VR teleoperation; **Planning & Control:** RRT*, FCL, state estimation, CBF-CLF QPs, MPC; **Learning:** PyTorch, ACT, DAgger, Diffusion Policy, PPO, flow matching; **Simulation:** Isaac Lab/Sim, MuJoCo, Drake; **Software:** Python, C++, Linux, Docker, git, CI; **Robots:** Fairino FR3, xArm7, UR3e, UR5, Robotiq Hand-E, RealHand A7 (Bimanual Arm-Hand System), RealHand L6, LEAP Hand v1, Unitree G1 (simulation).

Work Experience

Graduate Research Assistant | Aug. 2021 – Present
Institute for Systems Research (ISR), University of Maryland | College Park, MD

• Develop grasp-synthesis, state-estimation, motion-planning, and feedback-control methods for dexterous manipulation under contact uncertainty; implement Python/C++ and ROS 2 software for physical arm–hand systems.
• Trained PPO with GDF rewards in **1,024 Isaac Lab environments**: **64/64** pregrasp arrivals versus **0/64** with palm-distance rewards. Published distributional RL (**CDC 2025**): **448 goals** versus **329/239** across **100 MuJoCo episodes**, with similar constraint-violation costs.
• Developed VR teleoperation software for an xArm7-RealHand-L6 robotic manipulator with wrist-to-Cartesian-velocity mapping, relative-joint-position hand control, grasp primitives, thumb-abduction control, and hand-state reset; integrated MoveIt Servo with Dimensional OS (DimOS) for real-time collision detection, arm speed modulation, and velocity damping near kinematic boundaries.

Research Intern | Jun. – Aug. 2022
MATRIX Lab, USM at Southern Maryland | California, MD

• Identified a discrete-time Crazyflie model from flight data and implemented a finite-horizon optimal formation controller in Python and Crazyswarm; executed leader–follower formation flights with 10 physical Crazyflie quadrotors under localization uncertainty.

Robotics Engineer | Mar. 2020 – Feb. 2021
Kognitive Robotics | Lagos, Nigeria

• Designed power and sensor–actuator interfaces in Altium, assembled a mobile robot, and implemented and tested its ROS software on physical hardware.

Robotics Trainee | Jan. 2020 – Feb. 2021
Robotics & Artificial Intelligence Nigeria (RAIN) | Ibadan, Nigeria

• Built ROS-based visual-SLAM, state-estimation, and flight-control software for a quadrotor's onboard computer.

Projects

ParcelStow — Temporal Robustness in Imitation Learning (Paper, project page, Code, Data & Models) | *Isaac Lab, PyTorch* 2026
• Authored ParcelStow, an Isaac Lab benchmark analyzing temporal robustness in imitation learning; generated **970K** expert states (**297** demonstrations) and trained three imitation baselines. At 2× execution speed, ACT success fell to **53%** versus **84%** for the expert, with failures concentrated in high-precision insertion.

Grasp Distance Fields — Feedback Grasp Execution (Project, Paper) | *C++, Python, QP* 2026
• Implemented a CBF-CLF quadratic-program controller for feedback grasp execution; simulated experiments lifted **46 of 50** objects, with a median QP solve time of **0.09 ms**.

realhand_l6_ros2 — Dexterous-Hand ROS 2 Software (Code) | *C++, SocketCAN* 2026
• Implemented joint/tactile-state interfaces, contact-gated control, and MoveIt 2 configuration with mock and virtual-CAN CI tests. Verified communication, feedback, tactile acquisition, and trajectory execution on a physical RealHand L6.

Selected Publications

• **C. Enwerem et al.** "Does Imitation Learning Preserve Temporal Robustness in Dexterous Manipulation? An Expert–Learner Comparison Across Task Execution Speeds." *arXiv preprint*, 2026. arXiv:2609.01453, project page, code, data & models.
• **C. Enwerem et al.** "Grasp Execution Without a Planner: Configuration-Space Grasp Distance Fields with Certified Safety & Guaranteed Quality." *Preprint*, 2026. arXiv:2608.00600, project page.
• **C. Enwerem et al.** "Safety-Aware Reinforcement Learning for Control via Risk-Sensitive Action-Value Iteration and Quantile Regression." *CDC 2025*. arXiv:2506.06954.
• Additional first-author publications covering control, safe learning, and robotic manipulation: IROS 2026, CDC 2026, CDC 2025, CDC 2024; full publication list on Google Scholar.

Education

Ph.D., Electrical & Computer Engineering, University of Maryland, College Park, MD. Expected Fall 2026. Aug. 2021 – Present
B.Eng., Electrical Engineering, University of Nigeria, Nsukka. Highest Honors. GPA 3.84/4.0. Aug. 2018

Clinton Enwerem

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Dear Technical Recruiter,

Application: Robotics Research Engineer

I implement Python/C++ robotics software, including ROS 2 hardware interfaces, robot descriptions, controllers, motion planners, simulations, and automated tests, then validate the software on physical robots. I am an Electrical and Computer Engineering Ph.D. candidate at the University of Maryland, graduating in Fall 2026, with five years of work in dexterous manipulation, motion planning, feedback control, and robot learning.

realhand_l6_ros2 implements a C++ SocketCAN hardware interface and exports joint and tactile state. The package includes contact-gated control, URDF/SRDF, MoveIt 2, mock hardware, virtual-CAN tests, CI, and physical communication and sensing tests. kinematic_planner_ros2 implements RRT*, Informed RRT*, and FCL primitive, mesh, and self-collision queries for N-DOF URDF manipulators. Grasp Distance Fields enforce collision and grasp constraints with a CBF-CLF quadratic program, which solved in 0.09 ms in the reported manipulation experiments.

ParcelStow evaluates imitation policies in simulation. I trained DAgger, Diffusion Policy, and ACT on 297 expert demonstrations of a Unitree G1 and RealHand L6 insertion task. At twice nominal execution speed, expert success was 84% and ACT success was 53%, with 35 of ACT's 47 failures occurring during insertion. The public release contains the repository, data and models, and stage-level outcomes required to reproduce the comparison.

My resume is attached. You can reach me at **enwerem@umd.edu** or **301-416-9430**.

Sincerely,

Clinton Enwerem

October 14, 2026